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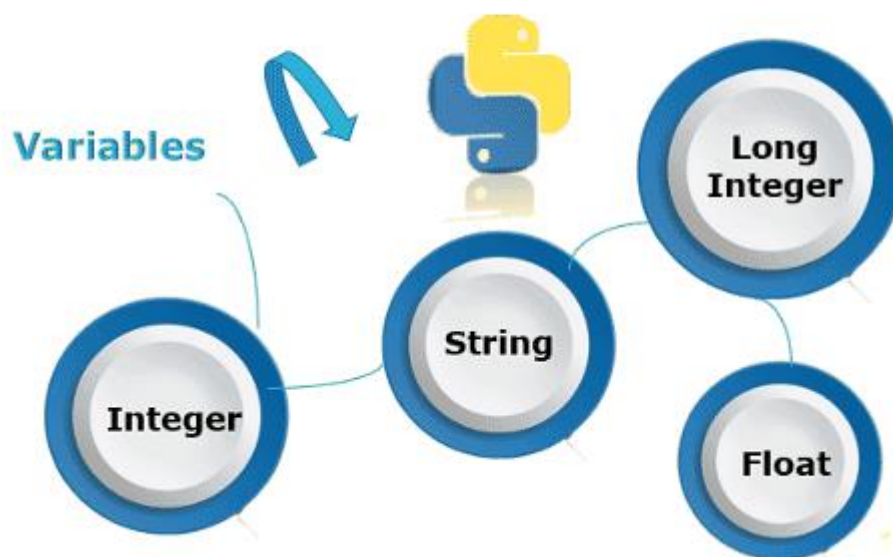
Variable

Python is not “statically typed”. We do not need to declare variables before using them or declare their type. A variable is created the moment we first assign a value to it. A variable is a name given to a memory location. It is the basic unit of storage in a program.

The value stored in a variable can be changed during program execution..

A variable is only a name given to a memory location, all the operations done on the variable effects that memory location.

Python Variables



Rules for creating variables in Python:

-



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- A variable name can only contain alpha-numeric characters and underscores (A-z, 0-9, and _).
- A variable name must start with a letter or the underscore character.
- A variable name cannot start with a number.
- Variable names are case-sensitive (name, Name and NAME are three different variables).
- The reserved words(keywords) cannot be used naming the variable.

Declaring Variable and Assigning Values:

- Python does not bind us to declare a variable before using it in the application. It allows us to create a variable at the required time.
- We don't need to declare explicitly variable in Python. When we assign any value to the variable, that variable is declared automatically.
- The equal (=) operator is used to assign value to a variable.

The multi-word keywords can be created by the following method:

- **Camel Case** - In the camel case, each word or abbreviation in the middle of begins with a capital letter. There is no intervention of whitespace. For example - nameOfStudent, valueOfVariable, etc.
- **Pascal Case** - It is the same as the Camel Case, but here the first word is also capital. For example - NameOfStudent, etc.
- **Snake Case** - In the snake case, Words are separated by the underscore. For example - name_of_student, etc.

Object Identity



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In Python, every created object identifies uniquely in Python. Python provides the guaranteed that no two objects will have the same identifier. The built-in id() function, is used to identify the object identifier.

Eg.

```
a = 50
b = a
print(id(a))
print(id(b))
# Reassigned variable a
a = 500
print(id(a))
```

Multiple Assignmen

Python allows us to assign a value to multiple variables in a single statement, which is also known as multiple assignments.

We can apply multiple assignments in two ways, either by assigning a single value to multiple variables or assigning multiple values to multiple variables..

1 . Assigning single value to multiple variables

Eg 1:

```
x=y=z=10
print(x)
print(y)
print(z)
```

2. Assigning multiple values to multiple variables:



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Eg 2:

```
a, b, c = 5, 10.5, "helloworld"  
print a  
print b  
print c
```

Can we use the same name for different types?

If we use the same name, the variable starts referring to a new value and type

```
a = 10  
a = "helloworld"  
  
print(a)
```

How does + operator work with variables?

```
a = 10  
b = 20  
print(a+b)  
  
a = "hw"  
b = "institute"  
print(a+b)
```

Can we use + for different types also?

```
a = 10  
b = "hw"  
print(a+b)
```

Python Variable Types



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There are two types of variables in Python –

- Local variable
- Global variable

Local Variable

Local variables are the variables that declared inside the function and have scope within the function. Let's understand the following example.

```
# Declaring a function
def add():
    # Defining local variables. They has scope only within a function
    a = 20
    b = 30
    c = a + b
    print("The sum is:", c)

# Calling a function
add()
```

Global Variable

Global variables can be used throughout the program, and its scope is in the entire program. We can use global variables inside or outside the function.

A variable declared outside the function is the global variable by default. Python provides the global keyword to use **global** variable inside the function. If we don't use the **global** keyword, the function treats it as a local variable. Let's understand the following example.



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```
# Declare a variable and initialize it
```

```
x = 101
```

```
# Global variable in function
```

```
def printFunction():
```

```
    # printing a global variable
```

```
    global x
```

```
    print(x)
```

```
    # modifying a global variable
```

```
    x = 'Welcome To Ssidigital'
```

```
    print(x)
```

```
printFunction()
```

```
print(x)
```

Delete a variable

We can delete the variable using the del keyword. The syntax is given below.

Syntax –

```
del < variable_name >
```

Example –

```
# Assigning a value to x
```

```
x = 6
```

```
print(x)
```

```
# deleting a variable.
```

```
del x
```

```
print(x)
```

Summary



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- Variables are referred to "envelop" or "buckets" where information can be maintained and referenced. Like any other programming language Python also uses a variable to store the information.
- Variables can be declared by any name or even alphabets like a, aa, abc, etc.
- Variables can be re-declared even after you have declared them for once
- In Python you cannot concatenate string with number directly, you need to declare them as a separate variable, and after that, you can concatenate number with string
- Declare local variable when you want to use it for current function
- Declare Global variable when you want to use the same variable for rest of the program
- To delete a variable, it uses keyword "del".



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